

Shaastra-2027

Elecfest: Signal Quest

1 Event Overview

Target Audience: 2nd, 3rd, and 4th year B.Tech students (ECE/EEE/EE).

2 Expected Sponsors

- ADI (Analog Devices, Inc) (Signal Quest 2026 sponsor)
- DigiKey
- KLA corporation
- Bharat Electronics Limited
- Texas instruments

3 Topics Covered

- Convolutions
- Properties of Linear systems
- Fourier series
- Fourier transform
- Laplace transform
- Z transform
- Sampling
- Analysis of basic filters

4 Round-1 Format Scheme

It will consist of an objective type quiz designed on the topics given above. There will be 35 MCQ, Numerical and comprehension type questions, to be completed in 90 minutes.

5 Round-1 Marking Scheme

- **Single Correct Type:** +4 for Correct Answer, -1 for Wrong Answer.
- **Multiple Correct Type:** +4 for Correct Answer, -2 for Wrong Answer. Partial Marks are distributed according to the number of chosen correct options.
- **Numerical Type:** +4 for Correct Answer, No marks will be deducted for Wrong Answer.
- **Comprehension Type:** +3 for Correct Answer, No marks will be deducted for Wrong Answer.
- **Subjective Type:** No subjective type questions will be included.

Selected participants will be selected for Round-2.

6 Round-2

- Round-2 will be offline.
- It will consist of questions including matlab tasks and signal generations tasks to be performed.
- Points will be awarded according to the difficulty of the problem.